Learned exchange-correlation functionals

Training-subset generation, pre-training, in-sample optimization and held-out generalization

2026-09-09

Scope

One network pair (exchange and correlation) per architecture and training-subset size, as separate cluster runs under one protocol, drawn once over every finished cell (the runs merged into one view). Each run carries a pre-training (the cloning of the parent), a fidelity certificate that gates its training array, and the array itself, one cell per (architecture, subset size).

Run	Architectures	Cells done / planned	Run directory
size	deep_3x16, deep_attn_3x16	deep_3x16 11/11, deep_attn_3x16 9/11	`run_20260902T145245Z`
families	deep0_3x16, deep0_attn_3x16, deep0_cusp_3x16	deep0_3x16 11/11, deep0_attn_3x16 2/11, deep0_cusp_3x16 0/11	`run_20260902T145247Z`
meta-GGA families	deep0_cusp_mgga_3x16, deep0_mgga_3x16, deep0_mgga_attn_3x16, deep0_rung35_mgga_3x16, deep0_rung35ms_mgga_3x16	0/55 (first certificate FAIL, four pending)	`run_20260902T145250Z`

- Subset sizes 1, 2, 3, 4, 5, 6, 7, 12, 15, 18, 26 points of the 26-point DFS pool [DFS21, SI Sec. II], chosen by Jensen-Shannon selection (Sec. 2).
- Basis 6-311++G(3df,2pd), grid level 3, density fitting, at every stage [cfg]. The validation-best checkpoint is read for every optimization figure.

1. Model: density variables and the DFS coordinates

With $n = n_\alpha + n_\beta$ and $\sigma = |\nabla n|^2$ [DFS21 Eqs. 4, 5]:

$$k_F = (3\pi^2 n)^{1/3}, \qquad s = \frac{|\nabla n|}{2k_F n}, \qquad r_s = \left(\frac{3}{4\pi n}\right)^{1/3},$$

$$\zeta = \frac{n_\alpha - n_\beta}{n}, \qquad \phi(\zeta) = \tfrac{1}{2}\left[(1+\zeta)^{4/3} + (1-\zeta)^{4/3}\right].$$

The networks read logarithmic compressions of these [DFS21 Eqs. 7 to 10; code: networks.py:621–657]:

$$x_0 = \ln\left(n^{1/3} + 10^{-5}\right), \qquad x_1 = \ln\phi(\zeta), \qquad x_s = \left(1 - e^{-s^2}\right)\ln(s+1), \qquad x_\alpha = \ln\frac{\alpha+1}{2}.$$

- Exchange input $[x_s, \ldots]$ with s of the doubled spin channel $2n_\sigma$ (the exchange network is evaluated per spin channel, next slides); correlation input $[x_0, x_1, x_s, \ldots]$ on the total density. The ellipsis holds the architecture's extra descriptors; on the meta-GGA rung the raw α is replaced by x_α.
- The 10^{-5} in x_0 is the DFS reference implementation's constant [dpyscf net.py:39; code: networks.py:27].

1. Model: the extra descriptors

Iso-orbital indicator (meta-GGA rung; SCAN's α [SCAN15 Eq. 2; DFS21 Eq. 6; code: metagga.py:8–9]), a contraction of the live density matrix P against the AO gradients, hence self-consistent and differentiable through the SCF:

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \qquad \tau = \tfrac{1}{2}\sum_i |\nabla\psi_i|^2 = \tfrac{1}{2}\sum_{\mu\nu} P_{\mu\nu}\, \nabla\chi_\mu \cdot \nabla\chi_\nu,$$

$$\tau_W = \frac{|\nabla n|^2}{8n}, \qquad \tau_{\text{unif}} = \tfrac{3}{10}(3\pi^2)^{2/3} n^{5/3},$$

with $\alpha \geq 0$ imposed by a smooth positive part rather than a hard clip [code: metagga.py:25–31].

Nuclear cusp pair (`cusp`), two bounded geometric features per grid point [code: descriptors.py:230–245]:

$$x_4 = \exp(-2Z_{\text{nearest}}\, r_{\text{min}}), \qquad x_5 = \tanh\left(\frac{1}{5}\ln\sum_A \frac{Z_A}{r_A}\right),$$

x_4 the density-form cusp envelope e^{-2Zr} [Steiner63, after Kato57], x_5 the log-compressed bare-nuclear potential. Both are geometric, so the synthetic pre-training mesh (Sec. 3) carries no values for them.

1. Model: the functional form

MLP output g at a grid point (GELU, `depth` $\times$ `nodes`) [code: networks.py:169, 379–404]:

$$F_x = 1 + \mathcal{L}_{1.804}(G_x \cdot g_x)\,, \qquad F_c = 1 + \mathcal{L}_2(G_c \cdot g_c)\,,$$

$$\mathcal{L}_\lambda(z) = \lambda\,\mathrm{sigmoid}(z - \ln(\lambda - 1)) - 1 \quad \text{[DFS21 Eq. 11]},$$

$$G = \tanh^2(s) \ \ \text{(GGA)}, \qquad G = x_s + \tanh^2(x_\alpha) \ \ \text{(meta-GGA) [DFS21 Eq. 12]}.$$

$\mathcal{L}_\lambda(0) = 0$ and $F \in (0, \lambda)$; $\lambda = 1.804 = 1 + \kappa_{\text{PBE}}$ for exchange [PBE96 Eq. 14] (DFS uses 1.174 [PRSB14]), $\lambda = 2$ so that $F_c \geq 0$ [DFS21 Eq. 13]. The gate G pins $F = 1$ where the gradient vanishes.

$$E_{xc} = \tfrac{1}{2}\left(E_x[2n_\alpha] + E_x[2n_\beta]\right) + E_c[n_\alpha, n_\beta] \quad \text{[OP79; code: descriptors.py:124]},$$

$$E_x[n] = \int n\,\epsilon_x^{\text{LDA}}(n)\,F_x\,d^3r, \qquad E_c = \int n\,\epsilon_c^{\text{PW92}}(r_s, \zeta)\,F_c\,d^3r \quad \text{[PW92]}.$$

Attention variants (`_attn`) [Vas17 Eq. 1; Xio20; code: net.py:55–136]: after the first hidden layer, the H hidden units are h tokens of dimension $d_k = H/h$ under pre-layer-norm self-attention across the token axis, a per-point re-mixing of hidden units:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \qquad x \leftarrow x + W^O\,\mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)(\mathrm{LN}(x)).$$

1. Model: the architectures

Architecture	Last layer	Rung, parent	Hidden	Heads	Extra
deep_3x16	Glorot	GGA, PBE	3 x 16	none	none
deep_attn_3x16	Glorot	GGA, PBE	3 x 16	4	none
deep0_3x16	zeroed	GGA, PBE	3 x 16	none	none
deep0_attn_3x16	zeroed	GGA, PBE	3 x 16	4	none
deep0_cusp_3x16	zeroed	GGA, PBE	3 x 16	none	cusp
deep0_cusp_mgga_3x16	zeroed	meta-GGA, SCAN	3 x 16	none	cusp, α

- Inputs: F_x from x_s, F_c from x_0, x_1, x_s; the cusp architectures add x_4, x_5 to both, the meta-GGA one x_α as well.
- deep_3x16 and deep0_3x16 have the same layers, inputs and parameter count; deep0_3x16 differs only in a zero-initialized final layer, $F_x = F_c = 1$ at initialization. After pre-training both are PBE clones differing only in their fitted weights: the two columns are replicates of one architecture under two starts; their combined held-out verdicts agree at every subset size, and on the energy leg alone they differ at 3 and 4 points (Sec. 5).

2. Subsets: the 26-point DFS pool [DFS21 SI Sec. II]

21 atomization energies (G2/97 molecules), 3 BH76 reaction energies, 2 IP13 ionization potentials; each point carries the species its loss term needs (an atomization its molecule and the H and Li anchors it contains, a reaction every reactant and product, an ionization the neutral and the cation); 30 unique species.

Idx	Point	Species (name, charge, spin)	Idx	Point	Species (name, charge, spin)
0	H2	H2 (0,0), H (0,1)	13	NO2	NO2 (0,1)
1	N2	N2 (0,0)	14	HN	HN (0,2), H (0,1)
2	FLi	FLi (0,0), Li (0,1)	15	O3	O3 (0,0)
3	CHN	CHN (0,0), H (0,1)	16	N2O	N2O (0,0)
4	CO2	CO2 (0,0)	17	CH3	CH3 (0,1), H (0,1)
5	F2	F2 (0,0)	18	CH2	CH2 (0,2), H (0,1)
6	C2H2	C2H2 (0,0), H (0,1)	19	H2O	H2O (0,0), H (0,1)
7	CO	CO (0,0)	20	H3N	H3N (0,0), H (0,1)
8	HLi	HLi (0,0), H (0,1), Li (0,1)	21	OH+N2_to_H+N2O (BH76)	HO (0,1), N2 (0,0), H (0,1), N2O (0,0)
9	Na2	Na2 (0,0)	22	OH+CH3_to_O+CH4 (BH76)	HO (0,1), CH3 (0,1), O (0,2), CH4 (0,0), H (0,1)
10	NO	NO (0,1)	23	HF+F_to_H+F2 (BH76)	HF (0,0), F (0,1), H (0,1), F2 (0,0)
11	CH	CH (0,1), H (0,1)	24	Li_IP (IP13)	Li (0,1), Li+ (1,0)
12	HO	HO (0,1), H (0,1)	25	C_IP (IP13)	C (0,2), C+ (1,1)

References: atomization energies [HK12] with H2O and C2H2 from [W411]; reaction energies [Goe17, BH76RC]; ionization potentials [NIST]. Code: dfs_pool.py, training_points.py:347–420.

2. Subsets: the descriptor distributions

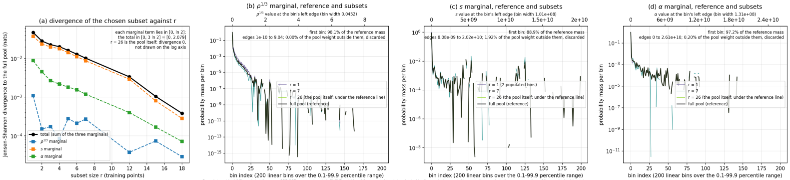
One PBE SCF per species (def2-SVP, grid level 1) [code: subset_selection.py:106–118, 448–470] gives at every grid point g with quadrature weight w_g

$$\rho^{1/3}, \qquad s = \frac{|\nabla\rho|}{2(3\pi^2)^{1/3}\rho^{4/3}} \ \text{[PBE96]}, \qquad \alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}} \ \text{[SCAN15], clipped to } [0, 100].$$

A point's sample is the concatenation of its species' grid points; the reference is the concatenation over all 26 points. Each marginal is binned into 200 equal-width bins between its 0.1 and 99.9 percentiles and normalized to a mass function:

$$h_x^{(b)} = \sum_{g\,:\,x_g \in \text{bin } b} w_g, \qquad P_x^{(b)} = \frac{h_x^{(b)}}{\sum_{b'} h_x^{(b')}}.$$

Training-subset selection: Jensen-Shannon divergence to the full DFS pool



Descriptors (rho^1/3, s, alpha) from one PBE SCF per species at def2-SVP, grid level 1, every grid point weighted by its quadrature weight; a point's sample is the concatenation of its species' grid points. Each marginal is binned into 200 linear bins over the reference's 0.1-99.9 percentile range (samples outside the range are discarded, not clipped into the end bins) and normalized to a probability mass function; the divergence is the equally weighted sum over the three marginals of 0.5 [KL(P||M) + KL(Q||M)], M = (P + Q) / 2, P the pool and Q the subset. For each r the subset is the exhaustive minimizer over all C(26, r) combinations.

2. Subsets: the metric and the search [Lin91]

The distance between a candidate subset S and the pool is the Jensen-Shannon divergence [code: subset_selection.py:213–246, 472] summed over the three marginals with equal weights, P_x the pool and Q_x the subset mass function, probabilities floored at 10^{-12}, natural logarithm:

$$\mathrm{JSD}(S) = \sum_{x \in \{\rho^{1/3}, s, \alpha\}} \tfrac{1}{2}\left[\mathrm{KL}(P_x \,\|\, M_x) + \mathrm{KL}(Q_x \,\|\, M_x)\right], \qquad M_x = \tfrac{1}{2}(P_x + Q_x),$$

$$\mathrm{KL}(P \,\|\, Q) = \sum_b P^{(b)} \ln \frac{P^{(b)}}{Q^{(b)}}, \qquad \text{each marginal term in } [0, \ln 2].$$

For each $r \in \{1, \ldots, 7, 12, 15, 18, 26\}$ the subset is the exhaustive minimizer over all $\binom{26}{r}$ combinations (9.7 million at $r = 12$); a subset with no grid point inside a marginal's range gets an infinite divergence.

- The linear edges reach the low-density tails (s to 2.0×10^{10}, α to 2.6×10^{10}, $\rho^{1/3}$ from 10^{-10} to 9.04), so the first bin holds 98.1, 88.9 and 97.2 percent of the weighted reference mass: the whole chemical range for s and α, the vacuum tail ($\rho < 9.2 \times 10^{-5}$) for $\rho^{1/3}$.
- The metric therefore compares the low-density tails of the molecular grids (s, α) and the valence and core region ($\rho^{1/3}$); the s term is 74 to 89 percent of every divergence.

2. Subsets: the chosen subsets [ledger subset_index_log.json; plot_subset_jsd.py]

r	Points (ledger index order)	AE / BH76 / IP	JSD	$\rho^{1/3}$	s	α
1	H2O	1 / 0 / 0	0.0475	0.00109	0.03764	0.00881
2	HLi, OH+N2_to_H+N2O	1 / 1 / 0	0.0284	0.00015	0.02372	0.00455
3	OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	0 / 2 / 1	0.0228	0.00017	0.01995	0.00267
4	CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	1 / 2 / 1	0.0203	0.00008	0.01801	0.00218
5	HLi, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	1 / 2 / 2	0.0163	0.00028	0.01418	0.00183
6	HLi, CH, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	2 / 2 / 2	0.0130	0.00021	0.01120	0.00156
7	CHN, CO2, HLi, CH, OH+N2_to_H+N2O, Li_IP, C_IP	4 / 1 / 2	0.0102	0.00027	0.00877	0.00120
12	CO2, C2H2, CO, HLi, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	7 / 3 / 2	0.0034	0.00004	0.00292	0.00040
15	FLi, CHN, CO2, C2H2, CO, HLi, NO, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	10 / 3 / 2	0.0010	0.00007	0.00081	0.00017
18	N2, FLi, CHN, CO2, C2H2, CO, HLi, Na2, NO, CH, NO2, N2O, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	13 / 3 / 2	0.0004	0.00003	0.00028	0.00007
26	the whole pool	21 / 3 / 2	0	0	0	0

- The $\rho^{1/3}$, s and α columns are the three marginal terms of the JSD (their sum is the total); every value is recomputed from the descriptor caches and agrees with the ledger to 10^{-9}. The subsets are nested only where the minimizer happens to nest; the 2-to-6-point subsets are barriers and ionizations with one or two molecules.

3. Pre-training: targets and objective

Each network is fitted pointwise [code: pretrain.py:261–341, pretrain_data_gen.py] to reproduce its parent (PBE on the GGA rung, SCAN on the meta-GGA) on the parent's own densities of the DFS pre-training inventory plus every atom of the BH76 and W4-11 pools (38 systems on the GGA rung, 36 on the meta-GGA; the certificate re-evaluates 39 and 38), with a synthetic (r_s, s, α) mesh [DFS21 SI] at 30 percent of the weight:

$$t_x = F_x^{\text{parent}} - 1 = \frac{\epsilon_x^{\text{parent}}}{\epsilon_x^{\text{LDA}}} - 1, \qquad t_c = F_c^{\text{parent}} - 1 = \frac{\epsilon_c^{\text{parent}}}{\epsilon_c^{\text{PW92}}} - 1,$$

the exchange rows posed per spin channel at the doubled density $(2n_\sigma, 4\sigma_{\sigma\sigma})$ [OP79]. The objective is the integration-weighted pointwise residual plus a per-system energy term:

$$\mathcal{L}_{\text{pre}} = \frac{\sum_i w_i \, (F_i^{\text{NN}} - 1 - t_i)^2}{\sum_i w_i} + w_E \, \frac{1}{N_{\text{sys}}} \sum_s \Big(\sum_{i \in s} w_i^{\text{grid}} \, n_i \, \epsilon_i^{\text{LDA}} \, F_i^{\text{NN}} - E_{xc,s}^{\text{parent}} \Big)^2,$$

$$w_i = |n_i \, \epsilon_i^{\text{LDA}}| \, w_i^{\text{grid}}, \qquad w_E = 0.1.$$

- The energy term is needed because the pointwise residual alone does not bound a system's energy.
- Adam, 20000 steps, learning rate 10^{-3} to 10^{-5} from the midpoint, clip 1.0, a 20 percent validation slice scored every 50 steps with patience 300 checks [cfg `pretrain:`].

3. Pre-training: the fidelity certificate

The saved networks are re-evaluated [code: cluster/fidelity.py, grid_config.py:287–312] on every system at the production basis and grid and compared with the parent:

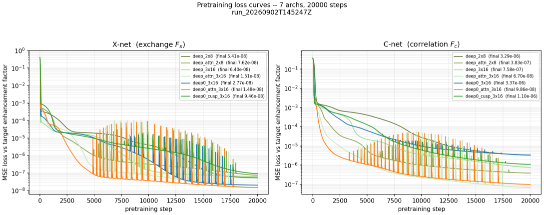
$$\Delta E_{xc,s} = E^{\mathrm{NN}}_{xc,s} - E^{\mathrm{parent}}_{xc,s}, \qquad \Delta AE = \Delta E_{xc,\mathrm{mol}} - \sum_A \Delta E_{xc,A},$$

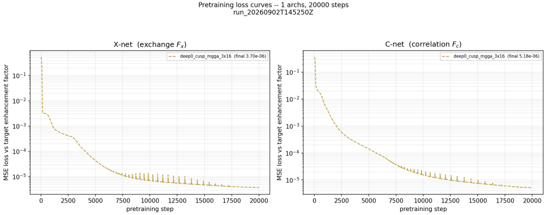
$$\max_{\mathrm{atoms}} |\Delta E_{xc,A}| \le 1.0 \text{ mHa}, \qquad \mathrm{mean}_{\mathrm{mol}} |\Delta AE| \le 1.0 \text{ kcal/mol}, \qquad \max_{\mathrm{mol}} |\Delta AE| \le 2.0 \text{ kcal/mol}.$$

A failing certificate holds the group's training array; every trained cell starts from a network that reproduces its parent to within these tolerances.

Architecture	Parent	Loss X / C	max atom (mHa)	mean $\lvert\Delta AE\rvert$	max $\lvert\Delta AE\rvert$	Verdict
deep_3x16	PBE	6.4e-8 / 7.6e-7	0.210	0.162	0.519	PASS
deep_attn_3x16	PBE	1.5e-8 / 6.7e-8	0.058	0.031	0.115	PASS
deep0_3x16	PBE	2.8e-8 / 3.4e-6	0.439	0.114	0.450	PASS
deep0_attn_3x16	PBE	1.5e-8 / 9.9e-8	0.123	0.042	0.149	PASS
deep0_cusp_3x16	PBE	9.5e-8 / 1.1e-6	0.439	0.188	1.071 (AlCl3)	PASS
deep0_cusp_mgga_3x16	SCAN	3.7e-6 / 5.2e-6	3.132	2.062	7.257 (14 species)	FAIL

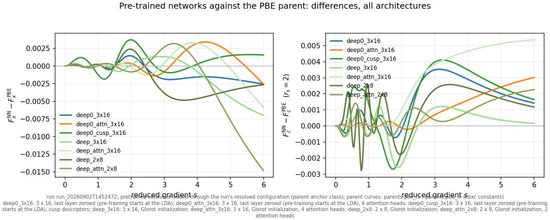
3. Pre-training: the loss curves

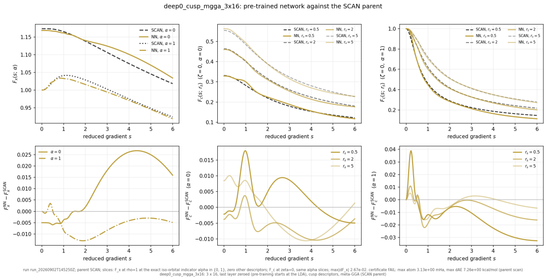




- Pointwise residual per step, exchange (left) and correlation (right); the recurring bursts from step 4000 to 8000 on are a property of the fit, not of the validation, which reloads nothing.
- GGA clones end at 1.5×10^{-8} to 9.5×10^{-8} (exchange) and 6.7×10^{-8} to 3.4×10^{-6} (correlation); deep_attn_3x16 descends earliest and lowest, at the longest fit (28 h against 3.8 h for its plain twin).
- The SCAN clone (right) stops 40 to 250 times higher in exchange and 1.5 to 80 times in correlation after the same 20000 steps and fails the certificate (previous slide). Its four siblings are pending; the meta-GGA training array is held.

3. Pre-training: the pre-trained networks against their parents





- Left: the difference $F^{\mathrm{NN}} - F^{\mathrm{PBE}}$ of every GGA clone, exchange at $n = 1$ and correlation at $r_s = 2$, along s. Within 10^{-3} for $s < 2.5$, at most 0.007 at $s = 6$ (deep_3x16 exchange); the three zero-init clones within 4.2×10^{-3} everywhere.
- Right: the SCAN clone. $F_x(\alpha = 0)$ is 0.01 to 0.027 above SCAN for $s > 2.5$, $F_c(\alpha = 1)$ 0.0003 to 0.016 below it from $s = 2$ at $r_s = 2$ (0.005 to 0.033 at $r_s = 0.5$); same-sign offsets over wide s ranges do not cancel between a molecule and its atoms.

4. In-sample: the training objective [DFS21 Eqs. 15–17]

One optimizer step per training group per epoch (a reaction, or a molecule with its density and potential references); the group loss is the DFS weighting ($\lambda_{RE} = 1$, $\lambda_n = 20$), with E^{NN} the self-consistent energy of the $N = 3$ cycle SCF and t the cycle:

$$\mathcal{L} = 1 \cdot \mathcal{L}_{AE} + 1 \cdot \mathcal{L}_{\mathrm{rxn}} + 1 \cdot \mathcal{L}_{IP} + 1 \cdot \mathcal{L}_{v_{xc}} + 20 \cdot \mathcal{L}_{\rho},$$

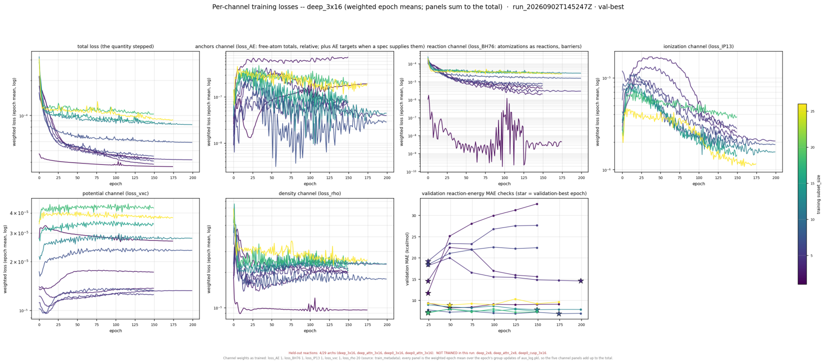
$$\mathcal{L}_{\mathrm{rxn}} = \frac{1}{|T|} \sum_{t \in T} w_t^2\, r_t^2, \qquad r_t = \sum_i c_i\, E_{i,t}^{\mathrm{NN}} - E^{\mathrm{ref}}, \qquad w_t = \left(\frac{t}{N-1}\right)^2,$$

$$\mathcal{L}_{AE} = 0.01\, \mathrm{mean}_Z\, \frac{(E_Z^{\mathrm{NN}} - E_Z^{\mathrm{anchor}})^2}{(E_Z^{\mathrm{anchor}})^2}, \qquad \mathcal{L}_{IP} = \mathrm{mean}\, (E_+^{\mathrm{NN}} - E_0^{\mathrm{NN}} - IP^{\mathrm{ref}})^2, \qquad \mathcal{L}_{v_{xc}} = \frac{\|V_{xc}^{\mathrm{NN}} - V_{xc}^{\mathrm{ref}}\|_F^2}{n_{\mathrm{AO}}^2},$$

$$\mathcal{L}_{\rho} = \mathrm{mean}_{\mathrm{mol}}\, \frac{\sum_g w_g\, (n_g^{\mathrm{NN}} - n_g^{\mathrm{ref}})^2}{N_e^2}, \qquad N_e = \sum_g w_g\, n_g^{\mathrm{ref}}.$$

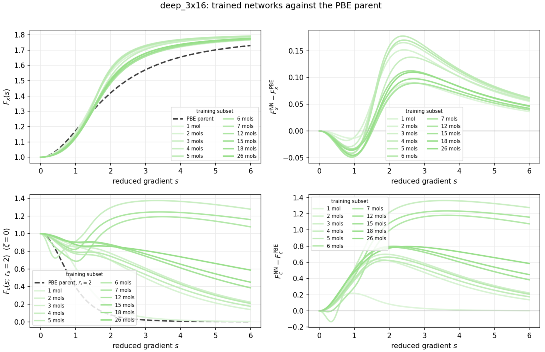
- $\mathcal{L}_{\mathrm{rxn}}$ is the DFS convergence-tail scoring ($w_t^2 = 0, 1/16, 1$ at $N = 3$) over the W4-11 atomizations, as reactions with the network's own atoms, and the BH76 barriers [W411, GMTKN55]; the free atoms are held near their exact energies [Chak93]; V_{xc} is compared on the reference density; the density against CCSD on the same grid, per electron. Code: losses.py:76, 366–396; train.py:1829–1835; oneshot.py:632–653.

4. In-sample: the training losses (deep_3x16)

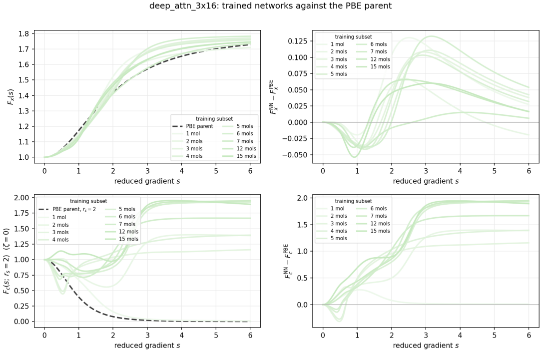


- SCF: 3 cycles from the converged PBE density, DFS mixing $D_{t+1} = (1 - a_t)D_t + a_t D_t^{\text{out}}$, $a_t = 0.3^t + 0.3$ [DFS21 SI], gradients through every cycle, the last mixed density scored. AdamW $10^{-3} \to 10^{-5}$ over the second half, weight decay 10^{-4}, clip 1.0, 200 epochs, validation MAE on 35 reactions every 25 epochs.
- The reaction channel falls in every cell, by 3x (deep0_3x16 at 18) to 190x (deep_attn_3x16 at 2); the density channel by 1.3x to 4.6x, both settled within 25 epochs. The potential channel, absent from the DFS objective, rises from 7 to 18 points in the plain networks (deep_3x16 at 18: 3.9e-5 to 4.3e-5) and falls in deep_attn_3x16 at 12 and 15.
- Validation MAE: the 2-to-6-point cells of deep_3x16 and deep0_3x16 rise after the first check (deep_3x16 at 2: 11.7 to 32.7 kcal/mol, best at epoch 25); the cells from seven points hold 6 to 11 kcal/mol; deep_attn_3x16's small cells fall (37 to 7 kcal/mol).

4. In-sample: the trained enhancement factors



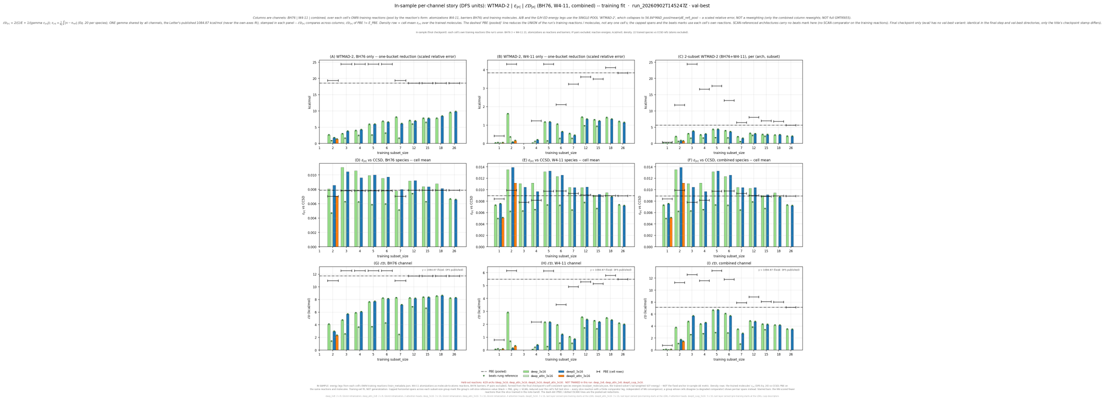
run run_20260902T145247Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at rho=1, zero extra descriptors; F_c at zeta=0, r_s=2; 11 completed cell(s); max|dF_x| 1.78e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_3x16: 3 x 16, Glorot initialization



run run_20260902T145247Z; checkpoint channel val_best (model_val_best.eqx); slices: F_x at rho=1, zero extra descriptors; F_c at zeta=0, r_s=2; 9 completed cell(s); max|dF_x| 1.32e-01; parent curves parents.pbe_fx / parents.pbe_fc (libxc constants).
deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads

- Exchange, every architecture and subset: the trained F_x is below PBE by 0.01 to 0.06 near $s = 0.8$ and above it from $s \approx 1.4$, by up to 0.18 (deep_3x16), 0.13 (deep_attn_3x16), 0.14 (deep0_3x16) at $s = 2.5$ to 3; the one-molecule cells of deep_3x16 and deep0_3x16 move least at $s = 0.8$.
- Correlation at $r_s = 2$: the one-molecule cells fall from 1 to 0.16 by $s = 2$ and to 0 beyond; the other cells stay between 0.45 and 1.3 up to $s = 2$ (PBE falls to 0.02 by $s = 3$), then split: deep_3x16 at 2, 3, 5, 6, 15, 18, 26 decays to 0.1 to 0.6 by $s = 6$ while 4, 7 and 12 rise to 1.1 to 1.3; every deep_attn_3x16 cell but the one-molecule one rises above 1 (up to 1.9 at $s = 6$).

4. In-sample: energy and density in DFS units on the training reactions



- Rows WTMAD-2 / ε_n / $\mathcal{ED}$ (Sec. 5 for the definitions), columns BH76 / W4-11 / combined; bars against PBE on each cell's own training reactions (caps), the dashed line PBE on their union (18.6, 3.83, 5.67 kcal/mol).
- All 33 cells are below PBE on $\mathcal{ED}$ over their own reactions: training WTMAD-2 0.00 to 4.4 kcal/mol against 5.67 on the union of the training reactions, deep_attn_3x16 0.05 to 2.45, deep0_attn_3x16 0.05 and 0.77. The grid-RMSE density of the trained molecules is below PBE in every cell (NN/PBE 0.72 to 0.97 deep_3x16, 0.57 to 0.76 deep_attn_3x16, 0.68 to 0.93 deep0_3x16); the per-electron ε_n of deep_3x16 is above the pooled PBE value from 2 to 18 points.

5. Held-out: the set and the metrics [Goe17; DFS21 Eqs. 20, 21]

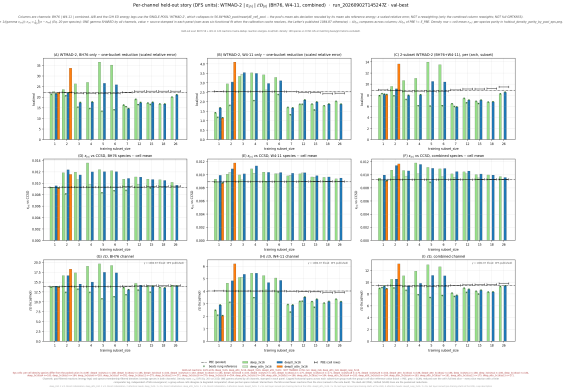
Test set: GMTKN55 BH76 barriers and W4-11 atomization energies, minus the 35-reaction validation slice and each cell's training reactions: 61 BH76 rows (three listed twice by name, four more as permuted-name twins; 54 barrier identities are scored) and 120 W4-11 reactions, 189 species with CCSD densities; the tables' n rxn column counts name-deduplicated rows (58 BH76 at one point). PBE is evaluated on each cell's own slice; the functionals with the same 3-cycle SCF they trained with, PBE's density converged.

$$\text{WTMAD-2} = \frac{56.84}{N} \sum_{i \in \{\text{BH76}, \text{W4-11}\}} N_i \frac{\text{MAD}_i}{\overline{|\Delta E_{\text{ref}}|}_i}, \qquad \varepsilon_n = \frac{1}{N_e} \int |n - n^{\text{ref}}| \, d^3r, \qquad N_e = \int n^{\text{ref}} \, d^3r,$$

$$\mathcal{ED} = \frac{2}{1/E + 1/(\gamma D)}, \qquad D_{\text{RMSE}} = \sqrt{\sum_g w_g (n_g - n_g^{\text{ref}})^2 \Big/ \sum_g w_g}.$$

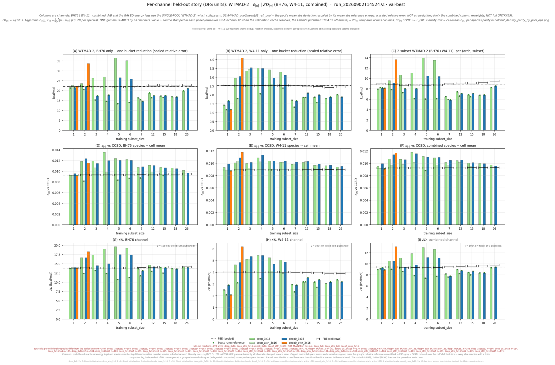
- WTMAD-2 over the two present subsets; alone, a pool reads $56.84\,\text{MAD}_i/\overline{|\Delta E_{\text{ref}}|}_i$, a scaled relative error. $\gamma = 1084.87$ kcal/mol on ε_n (DFS units) or $\gamma = E_{\text{PBE}}/D_{\text{PBE}}$ on the grid RMSE (self-calibrated).
- A cell beats PBE when its $\mathcal{ED}$ is below PBE's on the same slice. Pooled PBE: 8.92 kcal/mol WTMAD-2, ε_n 0.00923.
- The harmonic mean is set by its smaller leg: with $\gamma\varepsilon_n \approx 10$ kcal/mol here, 1 kcal/mol of E moves $\mathcal{ED}$ by 0.4 to 0.7, the whole density spread by at most 1.5..

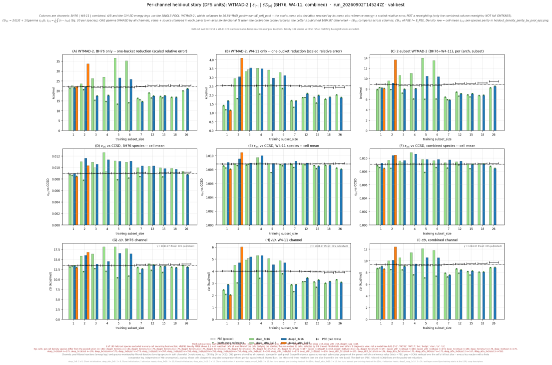
5. Held-out: energy and density in DFS units (full set)



- Energy (combined WTMAD-2, PBE 8.92): deep_3x16 7.92 at one point, 9.61 to 13.97 at two to six, 6.44 to 7.41 from seven to eighteen, 8.23 at twenty-six; deep_attn_3x16 5.94 to 8.26 over its nine cells, below PBE at every one; deep0_3x16 5.90 to 8.52, below PBE at 1, 3, 4 and 7 to 26, above at 2, 5, 6; deep0_attn_3x16 8.11 at one point, 13.64 at two.
- Density (ε_n, PBE 0.00923): at or above PBE in every cell but deep_attn_3x16 at five points (0.00883) and deep0_attn_3x16 at one (0.00906), by 0.2 to 28 percent.
- $\mathcal{ED}$: deep_3x16 and deep0_3x16 below PBE at 1, 7, 12, 15, 18, 26 (the two starts agree at every size); deep_attn_3x16 at every cell (lowest 7.40 to 7.73 at 5 to 7 points against 9.32 to 9.49); deep0_attn_3x16 at one point.

5. Held-out: the full set beside the set without the recurring density tail





- Left, the full set (the previous slide’s figure); right, the same figure with the recurring density tail removed from the density leg.
- The density tail recurs: over the 33 cells, bn (the largest ratio in 32 of the 33 cells), rkt17, cloo, bn3pi, cn, h2, rkt06 and cs are above 1.5x PBE in at least half of the cells carrying them. Without them (right) the deep_3x16 column from seven points is within 5 percent of PBE’s ε_n, deep_attn_3x16 below it at 1, 2, 5, 6, 7, 12, 15, deep0_3x16 at 15 and 26, deep0_attn_3x16 at 1; the $\mathcal{ED}$ verdicts change only for deep0_3x16 at three points.
- The comparison is asymmetric by protocol (an unconverged network density against converged PBE); the converged-SCF re-evaluation in flight is the direct test.

5. Held-out: per-cell results, combined leg, DFS units [holdout_by_pool_3x3_eps.csv]

Arch	r	E NN	E PBE	eps NN	eps PBE	ED NN	ED PBE	ΔED
deep_3x16	1	7.92	8.92	0.00951	0.00923	8.96	9.46	**-0.50**
deep_3x16	2	9.61	8.92	0.01071	0.00923	10.52	9.44	+1.08
deep_3x16	3	10.67	8.92	0.01068	0.00923	11.11	9.32	+1.79
deep_3x16	4	11.06	8.92	0.01182	0.00923	11.88	9.34	+2.53
deep_3x16	5	13.97	8.92	0.01113	0.00923	12.96	9.32	+3.64
deep_3x16	6	13.51	8.92	0.01091	0.00923	12.62	9.34	+3.28
deep_3x16	7	6.44	8.92	0.01017	0.00923	8.13	9.49	**-1.35**
deep_3x16	12	7.41	8.92	0.01047	0.00923	8.97	9.56	**-0.59**
deep_3x16	15	6.89	8.92	0.01004	0.00923	8.44	9.60	**-1.16**
deep_3x16	18	6.74	8.92	0.00999	0.00923	8.31	9.62	**-1.31**
deep_3x16	26	8.23	8.92	0.00973	0.00923	9.25	9.80	**-0.55**
deep_attn_3x16	1	8.26	8.92	0.00925	0.00923	9.06	9.46	**-0.40**
deep_attn_3x16	2	7.87	8.92	0.00972	0.00923	9.01	9.44	**-0.43**
deep_attn_3x16	3	7.22	8.92	0.00981	0.00923	8.60	9.32	**-0.72**
deep_attn_3x16	4	6.09	8.92	0.01020	0.00923	7.85	9.34	**-1.49**
deep_attn_3x16	5	6.03	8.92	0.00883	0.00923	7.40	9.32	**-1.92**
deep_attn_3x16	6	6.09	8.92	0.00973	0.00923	7.73	9.34	**-1.61**
deep_attn_3x16	7	5.94	8.92	0.00963	0.00923	7.57	9.49	**-1.91**
deep_attn_3x16	12	6.61	8.92	0.01019	0.00923	8.27	9.56	**-1.29**
deep_attn_3x16	15	6.47	8.92	0.00956	0.00923	7.97	9.60	**-1.63**
deep0_3x16	1	8.18	8.92	0.00997	0.00923	9.32	9.46	**-0.14**
deep0_3x16	2	9.16	8.92	0.01143	0.00923	10.54	9.44	+1.10
deep0_3x16	3	7.99	8.92	0.01062	0.00923	9.44	9.32	+0.12
deep0_3x16	4	8.06	8.92	0.01160	0.00923	9.83	9.34	+0.48
deep0_3x16	5	10.48	8.92	0.01099	0.00923	11.16	9.32	+1.83
deep0_3x16	6	10.40	8.92	0.01097	0.00923	11.10	9.34	+1.76
deep0_3x16	7	5.90	8.92	0.01051	0.00923	7.77	9.49	**-1.72**
deep0_3x16	12	7.11	8.92	0.01058	0.00923	8.78	9.56	**-0.78**
deep0_3x16	15	7.10	8.92	0.01013	0.00923	8.63	9.60	**-0.97**
deep0_3x16	18	6.82	8.92	0.00996	0.00923	8.37	9.62	**-1.26**
deep0_3x16	26	8.52	8.92	0.00961	0.00923	9.38	9.80	**-0.42**
deep0_attn_3x16	1	8.11	8.92	0.00906	0.00923	8.89	9.46	**-0.57**
deep0_attn_3x16	2	13.64	8.92	0.01168	0.00923	13.14	9.44	+3.70

- E in kcal/mol (2-subset WTMAD-2), eps per electron, ED under $\gamma = 1084.87$ kcal/mol; E PBE and eps PBE are the pooled values, ED PBE the cell-slice value. ΔED = ED NN $-$ ED PBE: **bold, negative** marks a cell below PBE (22 of 33).
- Verdicts within about 5 percent of PBE depend on the leg and the units: under the plain-MAE leg deep_attn_3x16 at 3, 5 and 6 flip; under self-calibrated grid-RMSE units deep_attn_3x16 at 2 and deep0_3x16 at 3 flip (holdout_ed_combined.csv). The deep_3x16 and deep0_attn_3x16 verdicts are the same under every choice.

6. Pending work

- Cells still queued: deep_attn_3x16 at 18 and 26, deep0_attn_3x16 from 3 to 26, deep0_cusp_3x16 at every subset.
- Meta-GGA group: four certificates pending; the deep0_cusp_mgga_3x16 FAIL holds the training array. The choice between dropping the architecture, refitting with more capacity or a higher energy-term weight, and an anchored form follows the four results.
- Converged-SCF re-evaluation of the 33 finished cells (DIIS at 10^{-8} Ha) and the CCSD T1 diagnostics for a model-free multireference list: two cluster jobs pending.
- Two training arms on the deep_3x16 architecture at 7, 12, 15, 18 and 26 points: 25 SCF cycles with the current protocol (arm A) and the dpyscf-parity protocol (arm B: no convergence freeze, the executed first-mix index, the randomized atomic/PBE seed, Adam with the plateau schedule, the DFS L2 and channel weights). Submitted 2026-09-08, no results yet.
- The figure suite's in-sample atomization-energy panel stays out of the documents until it uses the network's own atoms (it now subtracts fixed exact atomic energies and reports the absolute-energy offset).

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code the repository: `xcquinox/alec/networks.py`, `descriptors.py`, `metagga.py`, `config.py`, `xcquinox/net.py`, `xcquinox/alec/subset_selection.py`, `dfs_pool.py`, `training_points.py`, `pretrain.py`, `cluster/fidelity.py`, `cluster/grid_config.py`, `losses.py`, `train.py`, `oneshot.py`, `solver.py`, `evaluation.py`, `notebooks/analysis/make_ablation_arch_figure.py`; the figures and CSVs under `notebooks/analysis/figures_dfs_step7_*`; [cfg] the run configurations `hpcjobs/configs/dfs_step7.dfs6311_grid3_v7*.yaml`.